

Can this test answer its question?

Learning intention

Plan and use a fair test of how card height affects shadow height.

Question: do taller cards make taller shadows?

A team changed height and distance.

Help repair its plan.

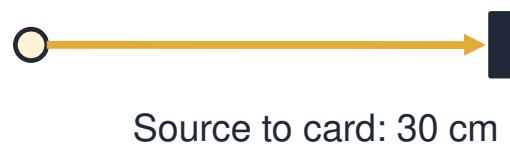
Recall our shadow studio

1. What blocks the light?
2. Can moving a card change shadow size?
3. What unit measures our shadow height?

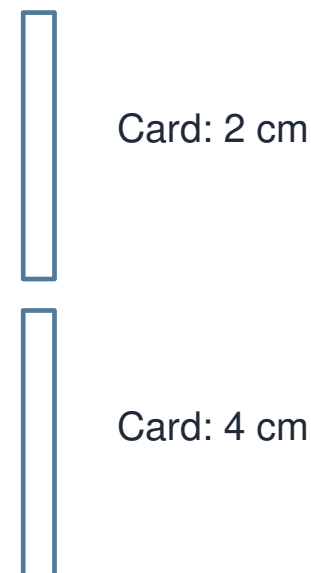
I spot two changed factors

PLAN NEEDS REPAIR

Test A



Test B



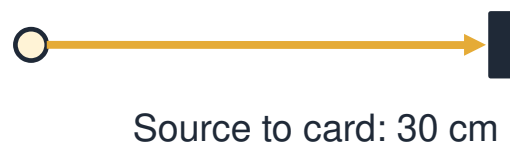
Same screen position: 60 cm from the source.

A uses a 2 cm-high card. B uses a 4 cm-high card. Their distances are different too.

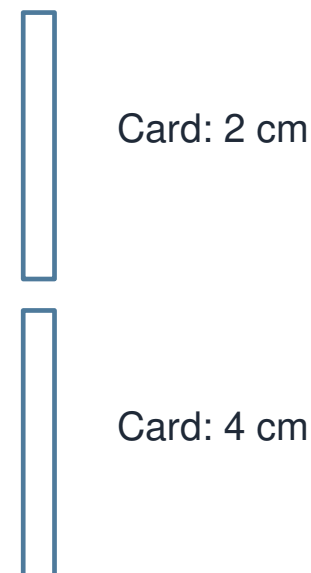
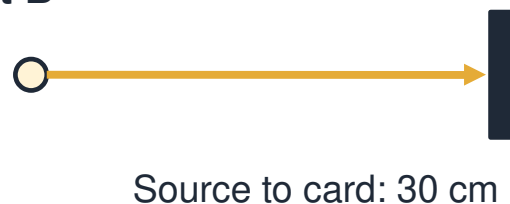
I repair the comparison

REPAIRED PLAN

Test A



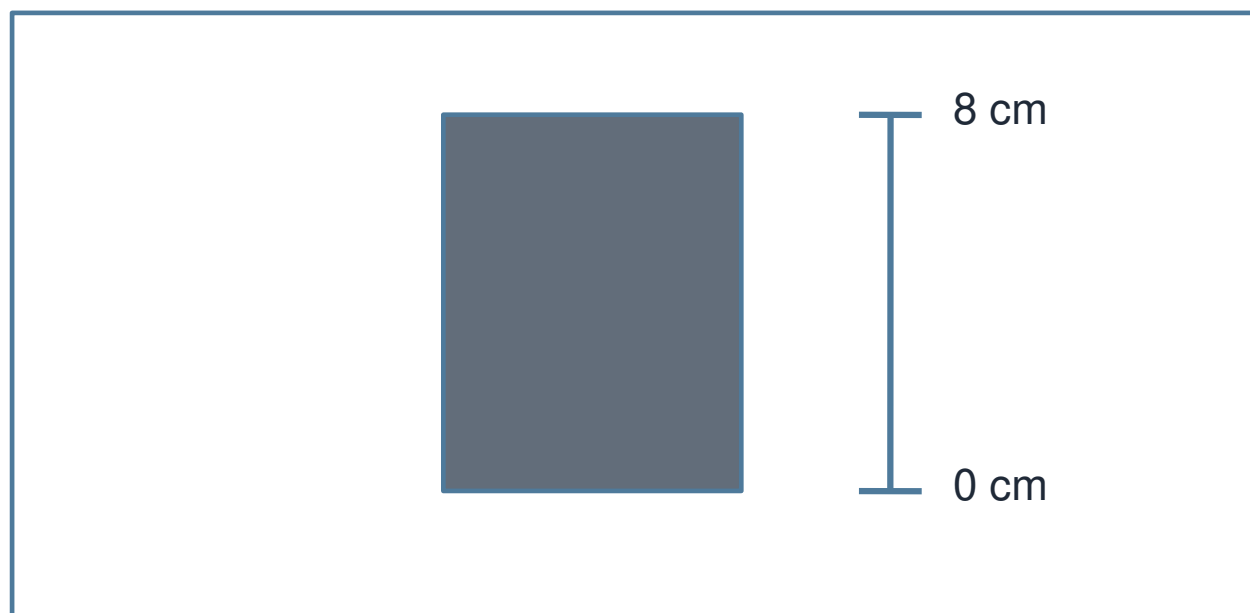
Test B



Same screen position: 60 cm from the source.

Change card height only. Keep source and screen still. Keep card position at 30 cm.

I measure and repeat consistently



Measure from the bottom edge to the top edge.

Measure bottom to top in centimetres. Use the same edge rule each time. Keep all readings.

Make a test contract

Choose the one change. Choose what to measure. Name two controls.

Plan part	Choose from these
Change	Card height / source-screen distance
Measure	Shadow height / card colour
Keep same	Card position / card height

Constructed teaching evidence

W1. Which choices answer our height question?

What should we do with 9?

Three readings include one difference. Keep the evidence. Check and repeat if useful.

Card height	Repeat 1	Repeat 2	Repeat 3
2 cm	4 cm	4 cm	4 cm
4 cm	8 cm	8 cm	9 cm
6 cm	12 cm	12 cm	12 cm

Constructed teaching readings, not class observations

W2. Can we keep a useful conclusion and keep the 9?

Which plan is fair for this question?

Question: how does card height affect shadow height?

- A** Change card height and source distance together.
- B** Change card height; keep the source, screen and card position fixed.
- C** Keep card height the same and only change its colour.

Choose your investigation route

Choose the test controls.

Write a method and conclusion.

Review the strength of a claim.

Use the evidence

Read or point to one clue.

Record what it supports.

Keep your explanation short.

Make your own science record

Complete your chosen route.

Use a diagram, words or speech.

Check one answer with the evidence.

Check and improve

Does my answer match the evidence?

Have I explained why?

Improve one label or sentence.

Lundy loop: improve our account

Choose how to share one idea.

Listen to another explanation.

Use evidence to improve our account.

What makes our conclusion stronger?

Three quick checks

E1. What factor did this test change?

E2. Name one thing kept the same.

E3. Should we delete a different reading just to fit the pattern?